

Coloured Surfaces and Kempe Surgery for D_5 -Flows: Structural Lemmas, a Reconfiguration Trap, and Finite Evidence

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Abstract

The standard five-cycle-double-cover conjecture asks whether every finite bridgeless graph has at most five Eulerian subgraphs covering each edge exactly twice. It remains unresolved. This note records a structural model and exact computations arising from an attempted proof.

For a loopless cubic graph, a five-cycle double cover is equivalently a conservative labeling of every edge by a two-subset of five coordinates, called a D_5 -flow. We show directly that such a flow canonically determines a closed triangulated surface whose faces are dual to the graph vertices and whose primal vertices are properly coloured with five colours. The coordinate circuits are precisely the primal vertex links, and

$$\chi(S(q)) = \sum_{i=0}^4 \kappa(C_i) - \frac{|V(G)|}{2}.$$

The usual componentwise coordinate transposition is a reversible triangle-band recolouring and regluing surgery. We give an exact permutation formula for its change in Euler characteristic and a cube example in which one switch changes a torus into a sphere.

We also prove a four-coordinate locking lemma and give a separately checkable 14-vertex contraction whose complete generalized-switch orbit has twelve normalized states, none satisfying the boundary condition needed to undo the contraction. This refutes a natural reconfiguration lemma, not the five-cycle-double-cover conjecture. A complementary Tait quotient proves that the analogous fully circuit-trapped branch is root-universal on cubic graphs, explaining why the degree-four trap does not lift unchanged. We also construct a planar cubic capped-ladder family proving that no constant number of root-component Kempe switches can always rescue two prescribed edges; this concerns a restricted strategy and does not refute rooted Kempe universality or FiveCDC. We prove that universal rooted feasibility on the smaller graph obtained by eliminating one edge of a minimum counterexample would imply FiveCDC; a still weaker sufficient premise only concerns independent roots in 3-edge-connected cubic graphs with the short-cycle geometry forced by edge elimination from girth ten. Finally, exhaustive computations show that all 33,598 terminal equal-Euler-characteristic Kempe plateaus on the 480 biconnected simple cubic graphs of order 14 are root-universal. A sharper fixed-distance descent test has 642,167 subplateaus and no failure. These are finite results only. The universal terminal-plateau statement is left explicitly as a conjecture.

1 Status, conventions, and contribution boundary

A k -cycle double cover, in the convention used here, is a list of at most k Eulerian edge-subsets such that every edge occurs in exactly two members. Empty members may be appended, so “at most five” and “five, allowing empty members” are equivalent. This agrees with the convention in Oum’s recent exposition: the general cycle-double-cover theorem gives eight Eulerian subgraphs, while the

reduction from eight to five remains Conjecture 18 [7]. Recent work on five-cycle double covers also continues to state the five-subgraph assertion as a conjecture and proves special cases [8, 9].

The orientable five-cycle-double-cover conjecture adds directed Eulerian orientations so that each edge is used once in each direction. It is a different, stronger statement; no orientation condition is imposed in this note.

The embedding aspect is established background, not a novelty claim. Catlin explicitly studied embedded graphs, facial colourings, and double cycle covers [2]; see also Zhang’s monograph [11] and Mohar and Thomassen’s text [10]. Ghanbari and Šámal formulate the cubic CDC in terms of embeddings without singular edges and study both approximation and twist operations [4, 5]. We have not completed a publication-grade prior-art search for the exact D_5 five-colour specialization, the component-switch Euler formula below, or the particular finite trap. Accordingly, this draft makes no categorical novelty claim. Its present value is that the statements, proofs, certificates, and limitations are exposed in a form that can be checked independently.

Literature-search TODO. Before circulation as a preprint, conduct a targeted search for prior work on unbounded Kempe-switch distance in two-edge-cut chains, especially prefix/suffix word obstructions for cycle-double-cover or nowhere-zero flow reconfiguration. The capped-ladder theorem below has not yet been compared adequately against that literature.

2 D_5 -flows

All graphs in the structural sections are finite and loopless. Parallel edges are permitted unless simplicity is stated. Edge objects, including parallel ones, are distinct.

Put

$$D_5 = \binom{\{0, 1, 2, 3, 4\}}{2} \subset \mathbb{F}_2^5.$$

We identify a pair ij with its weight-two incidence vector. A D_5 -flow on a graph G is a map

$$q : E(G) \longrightarrow D_5$$

such that

$$\bigoplus_{e \ni v} q(e) = 0 \tag{1}$$

at every vertex v . Since the graphs considered here are loopless, each incident edge appears once in (1).

Proposition 1 (cover-flow equivalence). *The D_5 -flows on G are in bijection with ordered five-tuples $(C_0, \dots, C_4)$ of Eulerian edge-subsets in which every edge occurs exactly twice.*

Proof. Given q , put

$$C_i = \{e \in E(G) : i \in q(e)\}.$$

The i -th coordinate of (1) says that C_i has even degree at every vertex. Every label has weight two, so every edge belongs to exactly two of the C_i .

Conversely, given the five Eulerian subsets, label an edge by the two indices of the members containing it. The label has weight two, and the five even-degree conditions are exactly (1). The constructions are inverse. $\square$

The following elementary local observation drives the surface model.

Lemma 2 (local coordinate triangle). *Let $a, b, c \in D_5$ and $a + b + c = 0$. Then there are three distinct coordinates i, j, k for which*

$$\{a, b, c\} = \{ij, ik, jk\}.$$

Proof. We have $c = a + b$. Since

$$\text{wt}(a + b) = 4 - 2|a \cap b| = 2,$$

the two pairs a, b meet in exactly one coordinate. Their union has three coordinates, and $c = a \triangle b$ is the third side of that coordinate triangle. $\square$

3 The coloured-surface dictionary

We first give a construction that does not assume an embedding in advance.

Theorem 3 (canonical coloured surface). *Let G be a connected loopless cubic graph with a D_5 -flow q . Then q canonically determines, up to colour-preserving cellular isomorphism, a connected closed triangulated surface $S(q)$ with the following properties.*

1. *The triangles of $S(q)$ are indexed by $V(G)$, and the dual graph is G .*
2. *The primal vertices are properly coloured with coordinates $0, \dots, 4$, in the sense that the three corners of every triangle have distinct colours.*
3. *If $C_i = \{e : i \in q(e)\}$, the components of C_i are in bijection with the primal vertices of colour i .*

Consequently, if $n = |V(G)|$, then

$$\chi(S(q)) = \sum_{i=0}^4 \kappa(C_i) - \frac{n}{2}. \quad (2)$$

Proof. For each $v \in V(G)$, Lemma 2 gives three coordinate colours on the incident labels. Take an abstract triangle Δ_v with those vertex colours, and assign the side opposite the remaining colour to the incident graph edge whose label is the two endpoint colours of that side.

For every graph edge $e = uv$, glue the side of Δ_u indexed by e to the side of Δ_v indexed by e , using the unique map that preserves its two endpoint colours. Every triangle side is paired exactly once. The quotient is therefore a closed two-dimensional pseudomanifold.

A quotient vertex can initially have more than one link component. Replace it by one vertex for each link component. Each resulting vertex has a circular link, so the normalized quotient is a closed triangulated surface. Splitting vertices does not change the triangles, their side pairings, or the dual graph. The surface is connected because its dual graph G is connected.

Fix a coordinate i . In a triangle having an i -coloured corner, the two sides at that corner are exactly the two incident graph edges whose labels contain i . Following the link of that corner through the side pairings therefore follows a component of C_i , and every such component arises this way. This proves (1)–(3).

The surface has n triangles and $3n/2$ primal edges. Its number of vertices is $\sum_i \kappa(C_i)$. Euler's formula now gives

$$\chi(S(q)) = \sum_i \kappa(C_i) - \frac{3n}{2} + n,$$

which is (2). $\square$

For a precise converse, call a face-colouring of a cellular embedding of a cubic graph *corner-proper* when the three face incidences at each graph vertex have three distinct colours. This definition rules out a single face occupying two corners at one vertex.

Proposition 4 (converse). *A corner-proper five-colouring of the faces of a cellular embedding of a connected cubic graph gives a D_5 -flow: label an edge by the colours of its two incident face occurrences. This construction and Theorem 3 are inverse after normalizing quotient vertices by link components.*

Proof. At a cubic vertex, let the three incident face colours be i, j, k . The three incident edge labels are ij, ik, jk , whose xor is zero. Thus the labels form a D_5 -flow. Taking the dual triangle at every graph vertex recovers the original side pairings and face-colour occurrences. Normalization separates exactly the distinct link components represented by distinct facial occurrences. $\square$

Remark 5. Propositions 1 and 4 show that the standard, unoriented five-cycle-double-cover problem for cubic graphs can be read as a problem of finding a corner-proper five-face-coloured cellular embedding. This is not an orientability statement: $S(q)$ may be orientable or nonorientable.

4 Bichromatic boundary curves and Kempe surgery

For $i \neq j$, define

$$Y_{ij} = C_i \triangle C_j = \{e : |q(e) \cap \{i, j\}| = 1\}. \quad (3)$$

It is an even subgraph and, in a cubic graph, a disjoint union of circuits.

Proposition 6 (boundary interpretation). *In $S(q)$, let H_{ij} be the primal subgraph induced by the primal vertices of colours i, j . After the standard dual isotopy,*

$$\partial N(H_{ij}) = Y_{ij}, \quad (4)$$

where $N(H_{ij})$ is a sufficiently small regular neighbourhood.

Proof. Check one primal triangle. If it contains neither i nor j , the neighbourhood has no boundary arc there. If it contains exactly one of them, the boundary arc around that corner crosses the two triangle sides whose labels contain exactly one of i, j . If it contains both, the boundary arc around their common primal edge crosses the other two sides, again exactly those whose labels contain one of i, j . These local arcs join across paired triangle sides and give (4). $\square$

Let K be a circuit component of Y_{ij} . A *component Kempe switch* transposes i, j in the label of every edge of K .

Lemma 7 (validity of a component switch). *A component Kempe switch sends a D_5 -flow to a D_5 -flow and is an involution.*

Proof. For an active label, transposing i, j adds the vector ij . At each graph vertex the circuit K has degree zero or two, so the total added vector in the local xor equation is zero. The switched labels still have weight two. Applying the same transposition a second time restores the original state. $\square$

On the coloured surface, this move has a concrete description. Recolour the triangles traversed by K by the transposition $i \leftrightarrow j$. Active side gluings still match because both incident triangles are recoloured. At an inactive ij -side incident with exactly one recoloured triangle, compose the side gluing with the endpoint flip. The result is exactly the coloured quotient for the switched flow. Thus the move is a reversible triangle-band recolouring and regluing surgery. It need not act on one fixed topological surface.

4.1 An exact Euler-change formula

Fix i, j , and abbreviate $A = C_i$, $B = C_j$. At each vertex of $A \cup B$, the nonzero membership patterns in (A, B) are

$$10, \quad 01, \quad 11.$$

Discard components of $A \cup B$ that are pure circuits. In every remaining component suppress the degree-two paths. The result is a cubic multigraph T_{ij} whose three edge classes 10, 01, 11 are perfect matchings. Write m_{10}, m_{01}, m_{11} for the associated fixed-point-free involutions on $V(T_{ij})$.

For a permutation π , let $z(\pi)$ denote its number of cycles, including fixed points.

Theorem 8 (one-switch Euler formula). *Let K be a component of Y_{ij} meeting T_{ij} , and let W be the vertices on the corresponding alternating circuit of $m_{10} \cup m_{01}$. Define*

$$\begin{array}{c|cc} & W & V(T_{ij}) \setminus W \\ \hline m'_{10} & m_{01} & m_{10} \\ m'_{01} & m_{10} & m_{01}, \end{array} \quad m'_{11} = m_{11}. \quad (5)$$

If q' is obtained by switching on K , then

$$\begin{aligned} \chi(S(q')) - \chi(S(q)) = \frac{1}{2} [& z(m'_{10}m_{11}) + z(m'_{01}m_{11}) \\ & - z(m_{10}m_{11}) - z(m_{01}m_{11})]. \end{aligned} \quad (6)$$

If K is a discarded pure circuit, the Euler change is zero.

Proof. Only $C_i = A$ and $C_j = B$ can change, so by (2) the Euler change is

$$\Delta\kappa(A) + \Delta\kappa(B). \quad (7)$$

The components of A that meet the cubic core correspond to the alternating circuits of the two matchings m_{10}, m_{11} ; similarly, the core components of B correspond to the alternating circuits of m_{01}, m_{11} .

If m_a, m_b are two perfect matchings, then on each alternating circuit of their union the product $m_a m_b$ has two permutation cycles. Therefore the number of alternating circuits is $\frac{1}{2}z(m_a m_b)$.

Switching i, j on K exchanges the 10- and 01-matching edges at exactly the vertices of W , giving (5). Substitute the before-and-after alternating-circuit counts into (7) to obtain (6). Pure circuit components outside the cubic core are unchanged or move whole from A to B ; their total contribution to (7) cancels. $\square$

4.2 A hand-checkable topology change

Take the cube with edge order

e	0	1	2	3	4	5	6	7	8	9	10	11
uv	01	03	04	12	17	23	26	35	45	47	56	67

and labels

$$01, 02, 12, 02, 12, 01, 12, 12, 01, 02, 02, 01. \quad (8)$$

The reader can check the xor equation at each of the eight vertices directly from the table. The coordinate component counts are

$$(\kappa(C_0), \dots, \kappa(C_4)) = (2, 1, 1, 0, 0),$$

so (2) gives $\chi = 0$. The edges $\{0, 1, 3, 5\}$ form one Y_{12} -circuit. Transposing 1, 2 on those four edges changes the component counts to

$$(2, 2, 2, 0, 0),$$

so the new connected surface has $\chi = 2$. A triangle-orientation propagation check shows that both surfaces are orientable. Hence this single legal switch changes a torus into a sphere. In particular, a global proof cannot treat all Kempe moves as curve slides on one fixed surface.

5 A four-coordinate locking mechanism

It is useful to distinguish component switches from a stronger move. For a coordinate pair $p = ij$, a *generalized factor switch* chooses any Eulerian edge-subset $Z \subseteq Y_{ij}$ and adds the vector ij to every label on Z . This remains a D_5 -flow by the same parity argument as Lemma 7.

Lemma 9 (four-coordinate one-step lock). *Suppose a D_5 -flow uses exactly the four coordinates $S = \{0, 1, 2, 3, 4\} \setminus \{t\}$, and suppose every nonempty coordinate class C_i , $i \in S$, is one circuit. From this state, no generalized factor switch can introduce t nontrivially modulo a global coordinate permutation.*

Proof. A switch with pair $p \subseteq S$ does not introduce t . For a pair $p = \{i, t\}$, the absence of t gives

$$Y_{it} = C_i.$$

A circuit has only two Eulerian edge-subsets, the empty set and the whole circuit. The empty switch does nothing. Switching all of C_i replaces i by t on every edge label containing i , which is exactly the global coordinate transposition $i \leftrightarrow t$. $\square$

The hypothesis of Lemma 9 need not be preserved by all switches among the used coordinates. Thus the lemma alone is a one-step statement, not an orbit theorem. The following finite example separately certifies orbit closure.

Proposition 10 (certified generalized-switch trap). *There is a simple 3-connected cubic graph G on 14 vertices and an edge e such that, after contracting e , the resulting degree-four graph has a generalized-factor-switch orbit with the following properties, modulo the global S_5 action:*

1. *the orbit has exactly 12 states;*
2. *every state uses four coordinates and each used coordinate class is one circuit; and*
3. *every state has side xor of weight four and therefore cannot be extended over the restored edge.*

The same contracted graph also has a different orbit containing extendible states. Therefore this is not a counterexample to FiveCDC.

Finite certificate. The parent graph has graph6 string

$$M?AA@BORD_CoEOAo?.$$

In lexicographic edge order, contract edge $15 = (3, 12)$. With the contracted edges indexed by the inherited order, the prescribed sides at the degree-four vertex are

$$U = \{5, 11\}, \quad V = \{15, 16\}.$$

One state in the orbit is

$$(01, 02, 03, 13, 03, 01, 23, 02, 03, 03, 02, 23, 23, 13, 12, 01, 23, 12, 01, 02). \quad (9)$$

Its local xors vanish, while both side xors are

$$q(5) + q(11) = q(15) + q(16) = 0123. \quad (10)$$

The companion file `orbit-certificate.txt` lists all twelve normalized rows and the complete normalized adjacency relation. A finite hand verification proceeds as follows. For each row, check the local xor equations and (10). For each coordinate pair, construct Y_{ij} , delete one edge from every circuit to obtain a spanning forest, and use the remaining edges as a binary cycle-space basis. Switch every basis sum and normalize under the 120 coordinate permutations. The resulting distinct rows are exactly the neighbours listed in the certificate. The adjacency list is connected, so the twelve rows form one complete orbit. Each row also directly satisfies the hypotheses of Lemma 9.

The supplied independent Python checker performs exactly these finite operations without a SAT solver, graph package, or canonical-labelling library. It also checks cubicity, bridgelessness, and 3-connectivity of the parent. $\square$

The proposition refutes the tempting assertion that every starting flow on a contraction can be repaired by generalized factor switches. It does not refute the existential assertion that some flow on the contraction extends; in fact, the other orbit contains 968 normalized extendible states.

5.1 Why the degree-four trap does not lift unchanged

The contraction trap raises a natural concern: perhaps a cubic Kempe orbit could remain locked in four coordinates with all four coordinate classes connected and still fail rooted universality. The next theorem excludes exactly that branch.

Call a component-Kempe orbit on a cubic graph *four-coordinate circuit-trapped* when every state uses exactly four coordinates and every used coordinate class is one circuit.

Theorem 11 (cubic D_4 trap theorem). *Every four-coordinate circuit-trapped component-Kempe orbit on a cubic graph is root-universal. More precisely, every pair of root edges is rescued in the starting state or after one component switch between used coordinates.*

Proof. Fix a state q , write S for its four used coordinates and t for the missing coordinate, and let r, s be distinct root edges. For clarity, globally rename S as $\{0, 1, 2, 3\}$.

Identify complementary labels in the three classes

$$\{01, 23\}, \quad \{02, 13\}, \quad \{03, 12\}. \quad (11)$$

These are three Tait colours. At each cubic vertex, the local xor triple is the triangle on three of the four coordinates, so its three labels lie in distinct classes in (11). Thus the quotient is a proper three-edge-colouring.

If $q(r) \cap q(s)$ contains a coordinate i , then both roots lie in

$$C_i = Y_{it}.$$

The circuit-trap hypothesis says that C_i is one circuit, so the roots are already rescued.

Otherwise the two root labels are disjoint. They are complementary two-subsets of S , hence have the same Tait colour A . Choose either other Tait colour B . The union of classes $A \cup B$ is a factor Y_p , where $\{p, S \setminus p\}$ is the third class. If r, s lie on the same component of Y_p , they are already rescued.

If they lie on different components, transpose the two coordinates of p on the component containing r . In the Tait quotient this interchanges the two active colours on that component: the colour of r changes from A to B , while the colour of s remains A . Labels from distinct complementary classes in (11) meet in exactly one coordinate, say k . The literal switch did not introduce t , so both roots now lie in $C'_k = Y'_{kt}$. The switched state remains in the assumed circuit-trapped orbit, and therefore C'_k is one circuit containing both roots. $\square$

The proof uses cubicity to obtain a proper Tait quotient and alternating bichromatic factor circuits. Neither fact survives at the degree-four contraction vertex. Theorem 11 therefore explains why Proposition 10 cannot be transplanted unchanged to a cubic counterexample. It closes only the fully circuit-trapped four-coordinate branch; a general rooted-orbit obstruction could still use five coordinates, a disconnected coordinate class, or transitions between those regimes.

6 Unbounded delay for root-component rescue

Fix distinct root edges r, s . Call a state *root-good* when some component of some Y_{ij} contains both roots. While the state is not root-good, a *root-component switch* means a component Kempe switch whose switched circuit contains exactly one root.

For $m \geq 1$, define a capped ladder L_m as follows. Its left cap has vertices a, b, p, q , its ladder has vertices u_i, v_i for $0 \leq i \leq m$, and its right cap has vertices c, d, x, y . The edge set is

$$\begin{aligned} &ap, aq, bp, bq, pq, \quad au_0, bv_0, \\ &u_i v_i \quad (0 \leq i \leq m), \\ &u_i u_{i+1}, v_i v_{i+1} \quad (0 \leq i < m), \\ &u_m c, v_m d, \quad cx, cy, dx, dy, xy. \end{aligned}$$

Thus the caps are copies of $K_4 - e$, and

$$|V(L_m)| = 2m + 10, \quad |E(L_m)| = 3m + 15.$$

The graphs are simple planar cubic graphs. They are bridgeless: cap edges lie on cap triangles, ladder edges lie on squares, and the attachment edges lie on cycles running through both rails and the caps.

Put

$$A = 01, \quad B = 02, \quad C = 12.$$

There is a Tait-supported D_5 -flow on L_m for which the labels on the successive two-edge cuts from left to right form the length- $(m + 2)$ prefix

$$A, C, B, A, C, B, \dots$$

Explicitly, label au_0, bv_0 by A , label $u_0 v_0$ by B , and assign the rail-pair and rung labels successively so that every ladder vertex sees A, B, C . On the left cap take

$$q(ap) = q(bq) = B, \quad q(aq) = q(bp) = C, \quad q(pq) = A,$$

and use the unique compatible Tait colouring on the right cap, up to exchanging its two inner vertices. Every vertex then sees A, B, C , so the local xor equations hold.

Theorem 12 (no constant root-component radius). *Choose one root edge inside each cap of L_m in the state above. Every root-component-switch sequence that reaches a root-good state has length at least*

$$\left\lfloor \frac{m+2}{3} \right\rfloor.$$

Consequently, no universal constant bounds the number of root-component switches required for rescue.

Proof. Let $D_0, \dots, D_{m+1}$ be the nested two-edge cuts in the displayed word. For an arbitrary D_5 -flow, the two edges of each D_j have the same label: xor the vertex equations on either side of the cut; internal edges cancel twice, leaving the xor of the two cut labels. Write the resulting cut-label word as $\lambda_0 \lambda_1 \dots \lambda_{m+1}$.

Let K be a factor component containing exactly one root. Connectedness and nestedness imply that the cuts met by K form a prefix when K contains the left root and a suffix when it contains the right root. Moreover, K is Eulerian, so it meets a two-edge cut in zero or two edges. A switch on K therefore applies one coordinate transposition to a prefix or suffix of the cut-label word.

After t switches, their at most t boundaries divide the initial word into at most $t+1$ intervals. On each interval the final labels are obtained from the initial labels by one fixed coordinate permutation, namely the ordered composition of the switches covering that interval.

If the final state is root-good through Y_h , its component containing both roots crosses every D_j . Hence

$$h \cdot \lambda_j = 1 \quad (0 \leq j \leq m+1),$$

with dot product over $\mathbb{F}_2$.

Partition the initial word into $\lfloor (m+2)/3 \rfloor$ disjoint consecutive triples A, C, B . No one triple can remain in a single interval. Otherwise one coordinate permutation π would act on all three entries, and

$$\pi(A) + \pi(C) + \pi(B) = \pi(A + C + B) = 0.$$

Dotting with h contradicts the preceding display, because the xor of three ones is one, not zero. Thus every disjoint triple contains a switch boundary in one of its two internal gaps. Those gaps are disjoint between triples, so one boundary splits at most one triple. Therefore

$$t \geq \left\lfloor \frac{m+2}{3} \right\rfloor,$$

as required. □

Remark 13 (scope). The proof deliberately relaxes the actual move system and uses only its necessary prefix/suffix action. It proves unbounded distance for the restricted root-component strategy. It does not produce a state that is unrescuable by root-component switches, does not refute unrestricted rooted Kempe universality, and does not prove or disprove FiveCDC. Literal breadth-first searches find finite rescue distances 3, 4, 5, 6, 8, 9 for selected roots in $L_1, \dots, L_6$, respectively. These computations are supplementary to, not inputs to, the proof.

7 Surface-Euler plateaus

Let $\mathcal{K}(G)$ be the state graph whose vertices are D_5 -flows modulo global coordinate permutations and whose edges are component Kempe switches. Put

$$\Phi(q) = \chi(S(q)).$$

For fixed graph edges r, s , call q (r, s) -good if some Y_{ij} -component contains both roots.

Definition 14. A Φ -plateau is a connected component after retaining only the equal- Φ edges of $\mathcal{K}(G)$. It is *terminal* when no state in it has a neighbour of larger Φ . A set of states is *root-universal* when, for every unordered edge pair r, s , it contains an (r, s) -good state.

Lemma 15 (connected-coordinate non-descent). *Suppose every C_i is one nonempty circuit. Then every component Kempe switch $q \mapsto q'$ satisfies*

$$\Phi(q') \geq \Phi(q).$$

If q lies in a terminal plateau, every such first switch is Φ -neutral.

Proof. Only the two switched coordinate classes, say C_i, C_j , can change. Initially their total component count is two. If both resulting classes are nonempty, each has at least one component and their total remains at least two.

Suppose instead that C'_i is empty. Then the switched component contains all of C_i , and $C_i \cap C_j = \emptyset$, since intersection edges are not in Y_{ij} . The two original circuits are also vertex-disjoint: two edge-disjoint even subgraphs of a cubic graph cannot both have degree two at the same vertex. The new C'_j therefore contains the two circuits as distinct components. Its component count is at least two, while C'_i contributes zero. The case $C'_j = \emptyset$ is symmetric. Thus

$$\kappa(C'_i) + \kappa(C'_j) \geq \kappa(C_i) + \kappa(C_j) = 2,$$

and (2) proves non-descent. Terminality excludes a strict increase. $\square$

Lemma 16. *Every state has a Φ -nondecreasing Kempe path to a terminal plateau.*

Proof. Move freely inside the current equal- Φ plateau. If it is not terminal, reach a state having a neighbour of strictly larger Φ and take that edge. The state graph is finite, so Φ can increase only finitely often. The process ends at a terminal plateau. $\square$

The computationally suggested statement is deliberately isolated.

Terminal-plateau conjecture. For every connected bridgeless cubic graph with a D_5 -flow, every terminal Φ -plateau is root-universal.

This statement is unproved. It is stronger than the finite observations below and is not asserted as a theorem.

7.1 Why the terminal conjecture would imply FiveCDC

The preceding conjecture is not merely a reconfiguration statement about graphs already known to have a D_5 -flow. Together with the standard minimum-counterexample reductions, it would prove FiveCDC.

Lemma 17 (two-root insertion equivalence). *Let H be cubic, let $r = u_1u_2$ and $s = v_1v_2$ be distinct edges, and obtain G by subdividing r, s with new vertices u, v and adding $e = uv$. Then H has an (r, s) -good D_5 -flow if and only if G has a D_5 -flow q in which u, v lie on the same component of $Y_{q(e)}(q)$.*

Proof. Choose a component K of $Y_{ij}(q)$ containing both roots. After subdividing r, s , give both halves their old labels, so K becomes a circuit K^* through u, v . Choose either u -to- v arc of K^* and transpose i, j on its edges. Every internal vertex sees either zero or two changed labels. At each of u, v , exactly one label changes by the vector ij . Label the new edge uv by ij ; this restores xor zero at both new vertices. All labels still have weight two. The new edge has label ij , so it is inactive in Y_{ij} ; transposition preserves activity, and hence the subdivided K remains a factor circuit through u, v .

Conversely, let q be a flow on G , put $P = q(e)$, and let K be the common $Y_P(q)$ -circuit through u, v . The edge e is inactive. At u , the other two labels therefore have the form $a, a + P$, and the same holds at v . Transpose the two coordinates of P on one u -to- v arc of K , and delete e . Internal vertex xors are unchanged, while the defect P at each endpoint cancels the deleted label. Exactly one of $a, a + P$ is transposed at each endpoint, so the two surviving labels become equal. Suppressing u, v therefore gives a D_5 -flow on H , and the suppressed image of K is one Y_P -circuit containing r, s . $\square$

Lemma 18 (bridgeless elimination). *Let G be a simple cyclically 4-edge-connected cubic graph of girth at least five. For $e = uv$, write*

$$N(u) - v = \{u_1, u_2\}, \quad N(v) - u = \{v_1, v_2\}.$$

Deleting u, v and adding $r = u_1u_2, s = v_1v_2$ produces a smaller simple 3-edge-connected cubic graph $G \div e$.

Proof. The elementary edge-reduction criterion stated immediately before Lemma 2.1 of [1] says that an edge of a connected cubic graph is irreducible only if it is a bridge, has an endpoint in a triangle not containing it, or has both endpoints on a 4-cycle not containing it. Thus the girth and connectivity hypotheses make $G \div e$ connected. The four neighbours displayed above are distinct, neither new edge is a loop or duplicates an old edge, and the two new edges are distinct; otherwise G has a triangle or a 4-cycle. Hence $G \div e$ is simple.

Suppose it has a bridge b , with shores X, Y . If $b = r$, then the endpoints of s lie on one shore while u_1, u_2 lie on opposite shores. Restoring u, v, e leaves the edge from u into the other shore as a bridge of G , a contradiction. The case $b = s$ is symmetric.

Otherwise b is an old edge. Neither r nor s crosses (X, Y) , so each of the pairs $\{u_1, u_2\}, \{v_1, v_2\}$ lies within one shore. If the two pairs lie on the same shore, b remains a bridge after restoring u, v, e . If they lie on opposite shores, $\{b, e\}$ is a 2-edge-cut of G . Both conclusions contradict 3-edge-connectivity.

It remains to exclude a 2-edge-cut $\delta_H(X)$ in $H = G \div e$. Let t be the number of roots crossing it. If $t = 2$, put the reinserted vertices u, v on the same shore: one spoke from each replaced root crosses and uv does not, giving a 2-edge-cut of G . If $t = 1$, put the vertex belonging to the noncrossing root on that root's shore and put the other new vertex there as well. One old cut edge and one spoke cross, again giving a 2-edge-cut of G . If $t = 0$ and both root pairs lie on one shore, put both new vertices there; the old 2-edge-cut persists. The only remaining placement has the two root pairs on opposite shores. Placing u, v on their respective shores then makes the two old cut edges together with uv a 3-edge-cut of G . Both shores contain circuits: a tree shore X of a 2-cut in a cubic graph would satisfy $3|X| = 2(|X| - 1) + 2$, which is impossible. This contradicts cyclic 4-edge-connectivity. Hence H is 3-edge-connected. $\square$

Proposition 19 (fixed-five minimum reduction). *If the standard Five-Cycle Double Cover Conjecture is false, it has a minimum counterexample that is simple, cubic, cyclically 4-edge-connected, and not 3-edge-colourable.*

Proof. Choose a counterexample with as few edges as possible. Components may be covered separately, padding each list with empty members and taking unions member by member. A loop may be deleted and then inserted into any two members, since a loop contributes degree two at its endpoint. Thus the counterexample is connected and loopless.

The usual minimum-counterexample proof for cycle double covers [11] preserves the fixed bound five. For completeness, here are the two reconstructions where that point matters. Across a 2-edge-cut, close the two shores by one artificial edge each. The two artificial edges receive labels in D_5 ; a global permutation of the five coordinates makes those labels equal, after which deleting the artificial edges and restoring the cut edges pastes the two D_5 -flows member by member. Hence a minimum counterexample is 3-edge-connected.

If a vertex has degree at least four, Fleischner's splitting lemma [3, Lemma III.26] supplies two incident edges whose split-off graph remains bridgeless. In a five-cover of that smaller graph, replace every occurrence of the artificial edge by the corresponding two-edge path through the split vertex. This changes no member index and preserves even degree. Minimality therefore makes the counterexample have maximum degree at most three. A degree-two vertex may likewise be suppressed: replace its two incident edges by one edge, then lift a cover by replacing every occurrence of the new edge by the original two-edge path. A connected loopless bridgeless graph has no degree-one vertex, so the minimum counterexample is cubic.

If a nontrivial 3-edge-cut remains, contract each shore in turn to a cubic vertex and cover the two smaller bridgeless graphs. At a cubic vertex the three D_5 -labels with xor zero are the three edges of a triangle on three coordinates. Every bijection between two ordered label triangles is induced by a permutation of the five coordinates. After applying such a permutation to one shore, the three boundary labels match and the covers paste without adding a member. Thus no nontrivial 3-edge-cut remains, which for a cubic graph is cyclic 4-edge-connectivity.

A proper 3-edge-colouring with colour classes M_1, M_2, M_3 gives the three-member double cover

$$M_1 \cup M_2, \quad M_1 \cup M_3, \quad M_2 \cup M_3;$$

append two empty members. The minimum counterexample is therefore not 3-edge-colourable. A loopless 3-edge-connected cubic multigraph with parallel edges is either separated by a 2-edge-cut or is the two-vertex graph with three parallel edges. The latter is 3-edge-colourable, so the minimum counterexample is simple. $\square$

Proposition 20 (conditional resolution). *The terminal-plateau conjecture implies the standard Five-Cycle Double Cover Conjecture.*

Proof. If FiveCDC is false, Proposition 19 gives a simple cyclically 4-edge-connected cubic minimum counterexample G . Huck proved that such a minimum FiveCDC counterexample has girth at least ten [6]. Choose any edge e . Lemma 18 gives a smaller bridgeless cubic graph $H = G \div e$. By minimality, H has a D_5 -flow.

Lemma 16 reaches a terminal plateau in its Kempe orbit. By the terminal-plateau conjecture, that plateau contains a state in which the two new edges r, s of H lie on one factor component. Lemma 17 extends this state over the inverse edge insertion, giving a D_5 -flow on G , a contradiction. $\square$

Remark 21 (a weaker sufficient premise). The proof needs neither every Kempe orbit nor a prescribed starting flow. It is enough to prove the following independent-pair feasibility statement:

Every simple 3-edge-connected cubic graph H with independent edges r, s , having girth at least eight, every 8-circuit through both roots, and every 9-circuit through at least

one root, has the following property: if H has a D_5 -flow, then it has one in which one factor component contains both roots.

Indeed, for $H = G \div e$ the new edges r, s are vertex-disjoint. A circuit of H using neither, one, or both new edges lifts to a circuit of G of the same length, one greater, or two greater, respectively. Thus $g(G) \geq 10$ implies $g(H) \geq 8$; moreover, every 8-circuit uses both roots and every 9-circuit uses at least one. Lemma 18 gives 3-edge-connectivity, and Lemma 17 finishes the same contradiction. This feasibility statement is still open.

7.2 Exact finite results

The state enumeration quotients only by the global S_5 action. Graphs were generated canonically by `geng -Cq -d3 -D3 n`. The complete order-12 terminal-plateau audit gives:

biconnected simple cubic graphs	81	
normalized D_5 -flows	25,960	
Kempe orbits	2,650	(12)
terminal Φ -plateaus	2,659	
non-root-universal terminal plateaus	0	

The terminal values range from -3 to 2 , and plateau sizes range from 1 to 432 . This is a complete finite census, not a proof of the terminal-plateau conjecture.

An additional support diagnostic found 420 terminal plateaus in which every state uses all five coordinates. Thus the order-12 positive result cannot be explained by always reaching a plateau with an unused coordinate and reducing to a four-coordinate argument. This support diagnostic is present in the producer report but is not among the fields recomputed by the focused independent verifier described below.

At order 14, the complete terminal-plateau computation gives:

biconnected simple cubic graphs	480	
normalized D_5 -flows	537,418	
terminal Φ -plateaus	33,598	(13)
non-root-universal terminal plateaus	0	

This includes terminal plateaus below the global maximum of an orbit.

There is a sharper finite descent test. For fixed roots r, s , let $d_q(r, s)$ be the minimum number of factor components in an intersection chain from r to s . Thus $d = 1$ is exactly root-goodness. Inside each terminal plateau, split the states of fixed distance $d > 1$ into components under neutral, distance-preserving moves. Every such component has a neutral exit to smaller distance:

terminal fixed- $d > 1$ subplateaus	642,167	
maximum distance	3	(14)
subplateaus without a neutral descent exit	0	

This suggests the following exact human target: every fixed- $d > 1$ subplateau in every terminal Φ -plateau has a neutral edge to smaller d . Repeated descent would prove terminal root-universality. The assertion is unproved outside the finite census.

As a separate unrestricted control, all 3,874 biconnected simple cubic graphs of order 16 were checked. Across 15,187,695 normalized flows, all 223,926,065 initially root-bad state/root pairs

were rescued using only switches on factor components containing exactly one root; the maximum minimum distance was five. No graph in that order-16 census has girth eight, so this computation does not directly reach the high-girth subclass in Remark 21. All of these statements are finite evidence only.

The first complete high-girth edge-elimination control uses a cubic graph on 130 vertices that has girth ten and is cyclically 4-edge-connected. For each of its 195 edges, the reduced 128-vertex rooted instance was solved independently. All 195 are root-good; their reduced girths are nine or ten. The complete positive labels are retained, and a separate checker reconstructs every reduction, enumerates the parent's 3-edge-cuts, verifies all D_5 equations, and checks factor connectivity. This is one graph, not a universal theorem.

7.3 Reproducibility and independent checks

The complete reports, producers, and focused independent verifiers are in the companion repository. From its root:

```
python3 scratch/audit_d5_surface_euler_switch.py
python3 scratch/check_d5_contraction_circuit_orbit_counterexample.py
python3 scratch/check_d4_cubic_trap_root_universality.py
python3 scratch/check_d5_capped_ladder_delay_family.py

python3 scratch/verify_d5_surface_chi_plateaus_order12.py
python3 scratch/verify_d5_root_euler_potential_reports.py
python3 scratch/verify_d5_root_kempe_order14_report.py
python3 scratch/verify_d5_terminal_chi_chain_lex_order14_report.py
python3 scratch/verify_d5_root_component_distance_order16_summary.py
python3 scratch/verify_d5_root_feasibility_lift13_girth10.py
```

The order-12 plateau verifier regenerates all graph identities and report arithmetic, then independently recomputes five deterministic boundary/extremal rows with a separately structured traversal. The order-14 fixed-distance verifier checks the frozen report digest, canonical graph order, per-row failure fields, and all summary arithmetic; its producer first reproduces the independent order-12 totals. The order-16 verifier checks the frozen graph identities, girth stratification, and summary totals. These report verifiers are independent semantic checks of selected structure and aggregate arithmetic, not second full enumerations; that limitation is part of the claim.

8 What has and has not been established

The human-checkable results in this note are:

1. the D_5 -flow/five-even-subgraph equivalence;
2. the canonical properly five-coloured surface construction and Euler formula;
3. the boundary interpretation of the bichromatic factors;
4. the exact one-switch Euler-change formula;
5. the four-coordinate one-step locking lemma and the cubic D_4 trap theorem;

6. the capped-ladder lower bound proving that root-component rescue has no universal constant radius; and
7. the fixed-five minimum-counterexample reduction, bridgeless edge elimination, and the rooted-feasibility implication to FiveCDC.

The cube topology change is a short explicit certificate. The 12-state contraction trap and the plateau tables are finite computer-assisted results with full instances, producers, and independent focused checkers.

No proof or disproof of the standard FiveCDC has been obtained. No orientable FiveCDC result has been obtained. The terminal-plateau conjecture is a proof target suggested by the computations, not a theorem. A responsible public submission would require independent checking by a graph theorist, a broader literature comparison, and archival publication of the exact code and reports.

AI-use disclosure

This project used OpenAI Codex agents extensively under human direction. The agents proposed the D_5 and coloured-surface viewpoints, derived candidate lemmas and proofs, wrote and ran the enumeration programs, searched for counterexamples to intermediate claims, produced the finite certificates, and drafted this manuscript. The human operator selected the research objective and directed the work. At the date of this draft, the manuscript has not received independent expert verification or peer review.

AI systems are not listed as authors. Any human who circulates or submits this note must personally verify the mathematical claims, take responsibility for the text and computations, confirm compliance with the venue’s AI policy, and replace the working-draft author line with the responsible author list. The disclosure is intentionally broad: it does not characterize the AI as a mere copy editor.

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